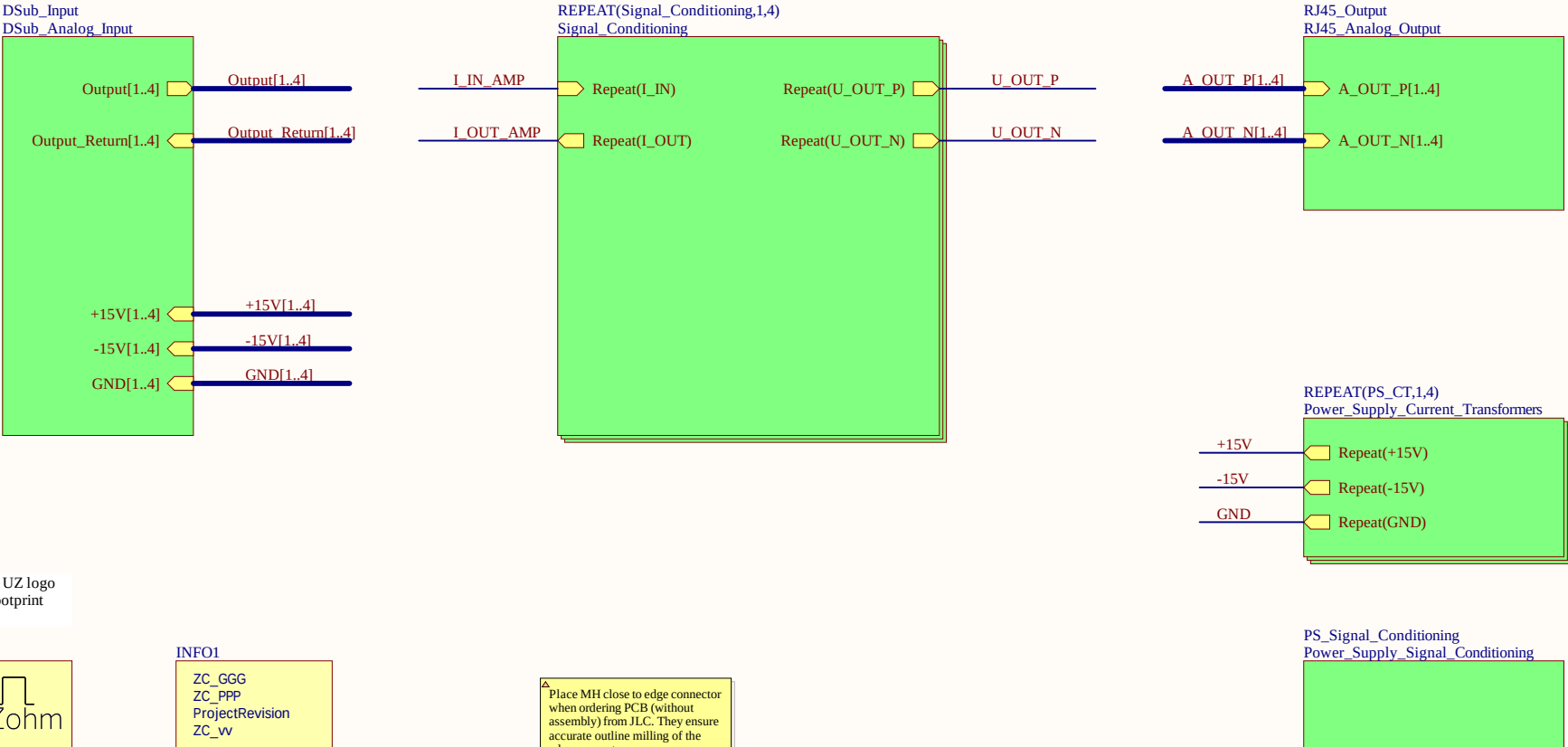


TopSheet

Output1	I_IN_AMP1
Output_Return1	I_OUT_AMP1
Output2	I_IN_AMP2
Output_Return2	I_OUT_AMP2
Output3	I_IN_AMP3
Output_Return3	I_OUT_AMP3
Output4	I_IN_AMP4
Output_Return4	I_OUT_AMP4

U_OUT_P1	A_OUT_P1
U_OUT_N1	A_OUT_N1
U_OUT_P2	A_OUT_P2
U_OUT_N2	A_OUT_N2
U_OUT_P3	A_OUT_P3
U_OUT_N3	A_OUT_N3
U_OUT_P4	A_OUT_P4
U_OUT_N4	A_OUT_N4



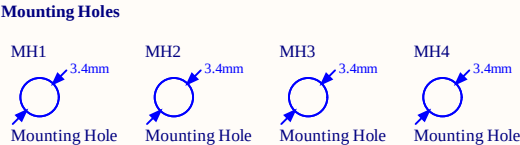
Chose size of UZ logo
by chosing footprint
type



INFO1
ZC_GGG
ZC_PPP
ProjectRevision
ZC_vv
ZC Serialnumber

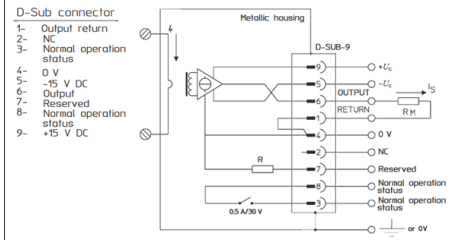
Place MH close to edge connector
when ordering PCB (without
assembly) from JLC. They ensure
accurate outline milling of the
edge connector.

INFO2
Project
ProjectRevision
AuthorParam
ProjectDate
Design Information



Title TopSheet.SchDoc		UltraZohm www.ultrazohm.com	
Revision: 01	Design Engineer: Tom Gaupp		
Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb			
Date: 27/05/2026 Sheet 1 of 12			

Connection



Signal Conditioning 1.4

Verstärkerschaltung Gain: $G_{amp} = 2$

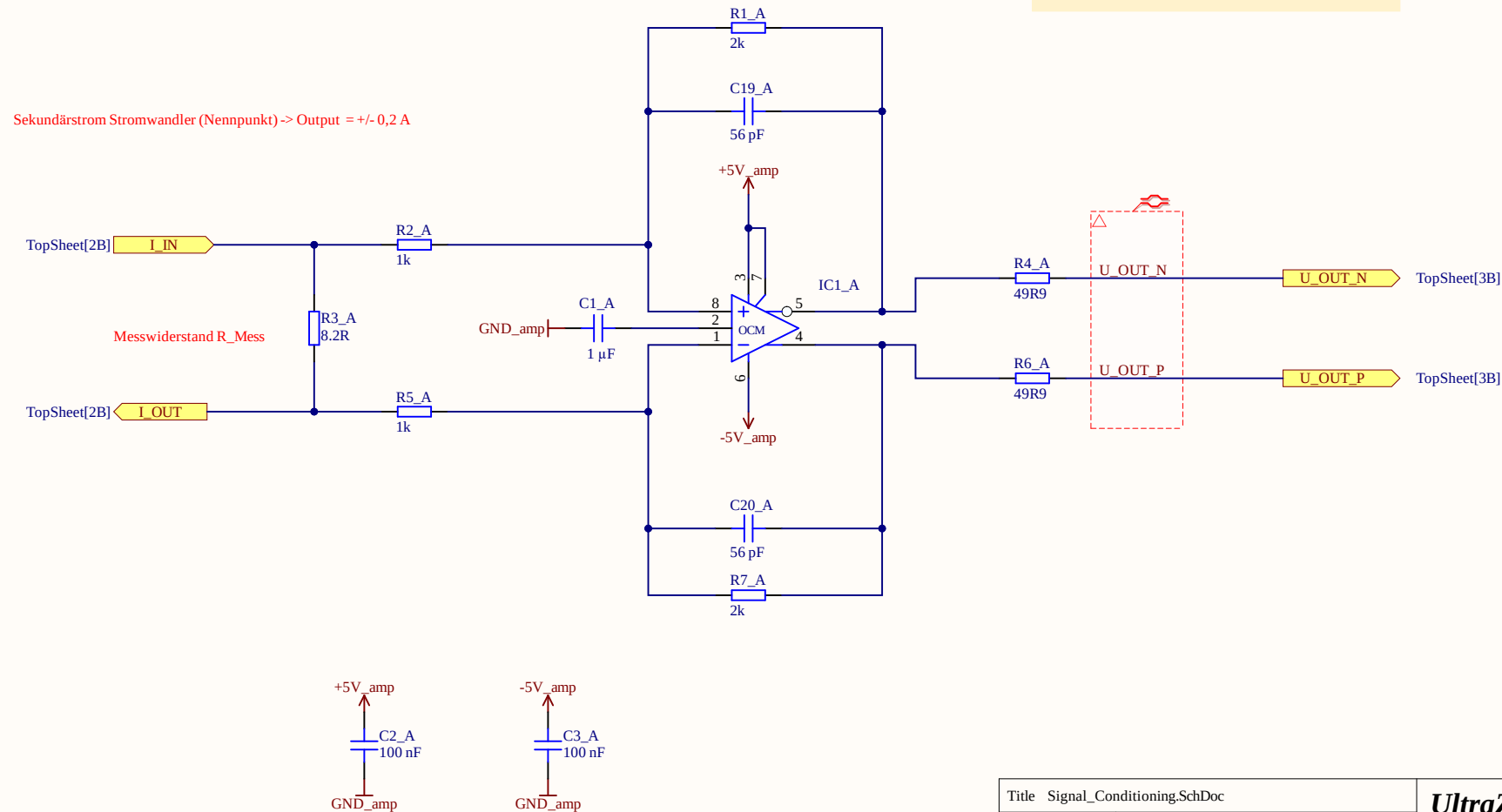
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

Rückkopplungskondensator: $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

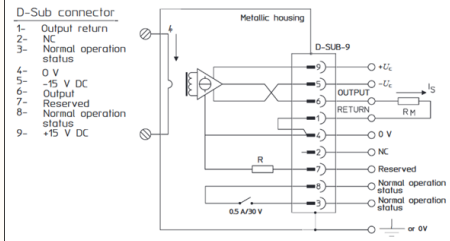
www.ultrazohm.com

Date: 27/05/2026

Sheet 2.1 of 12



Connection



Signal Conditioning 1.4

Verstärkerschaltung Gain: $G_{amp} = 2$

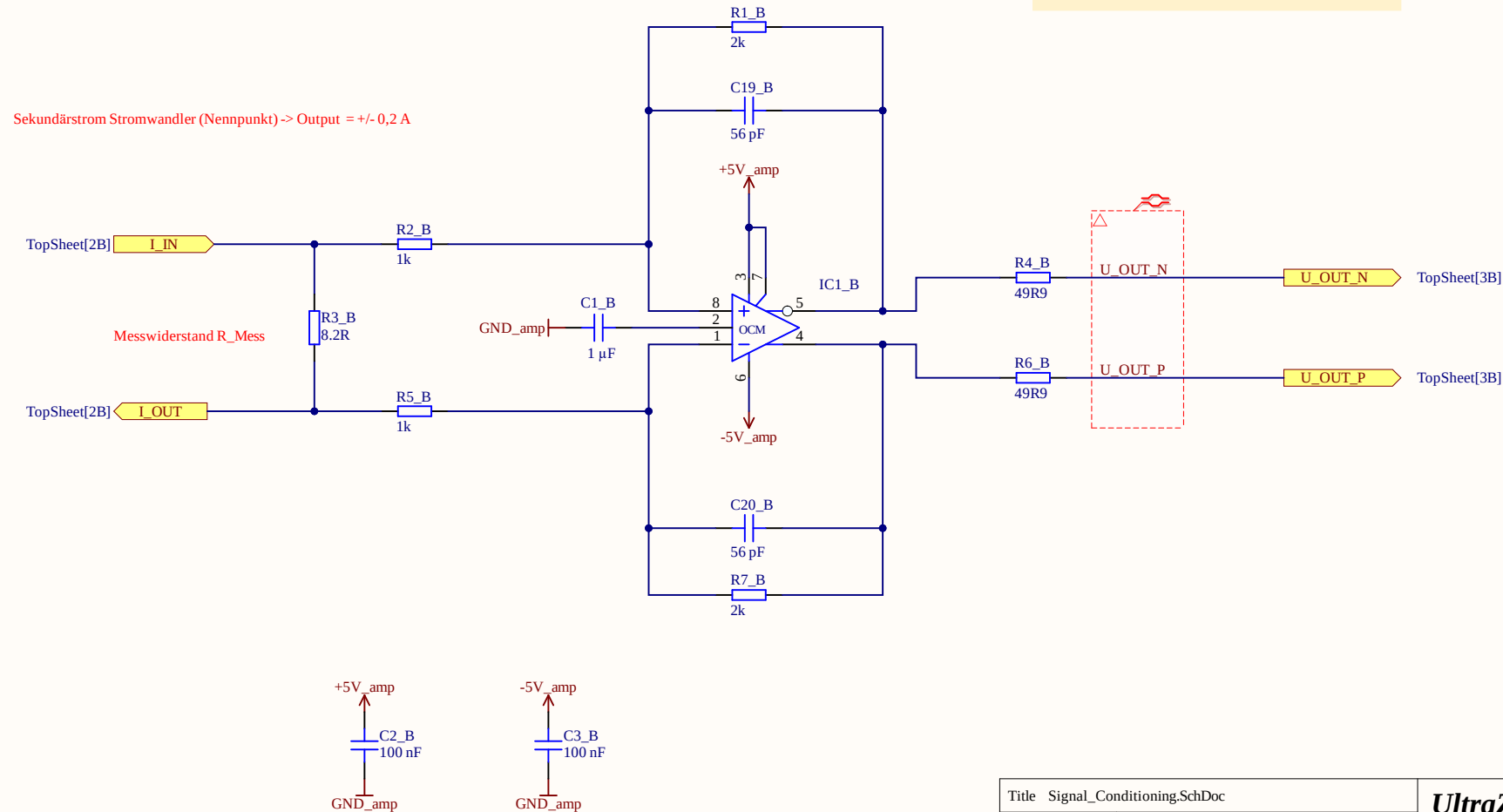
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

Rückkopplungskondensator: $C_f =$

$C_f = 1 / (2 \cdot \pi \cdot R_f \cdot f_g) = \dots =$



Title: Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

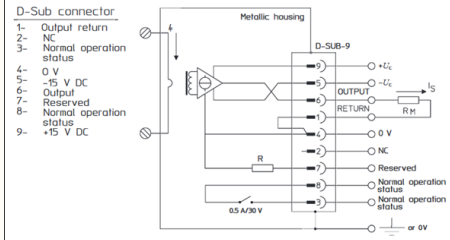
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Date: 27/05/2026

Sheet 2.2 of 12



Connection



Signal Conditioning 1..4

Verstärkerschaltung Gain: $G_{amp} = 2$

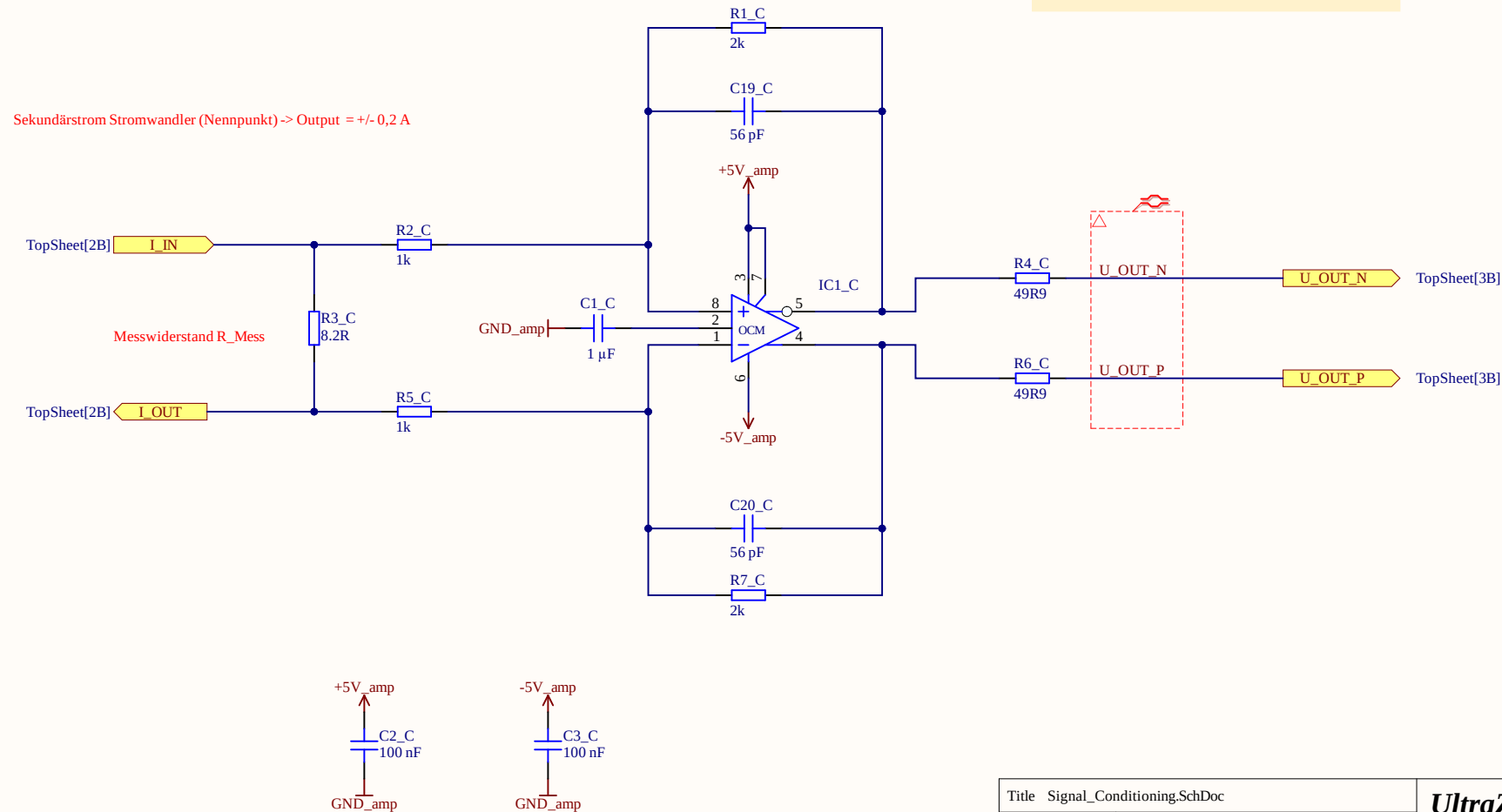
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

Rückkopplungskondensator: $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

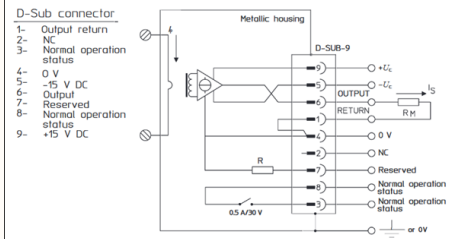
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Date: 27/05/2026

Sheet 2.3 of 12



Connection



Signal Conditioning 1.4

Verstärkerschaltung Gain: $G_{amp} = 2$

$$G_{amp} = R_F / R_G$$

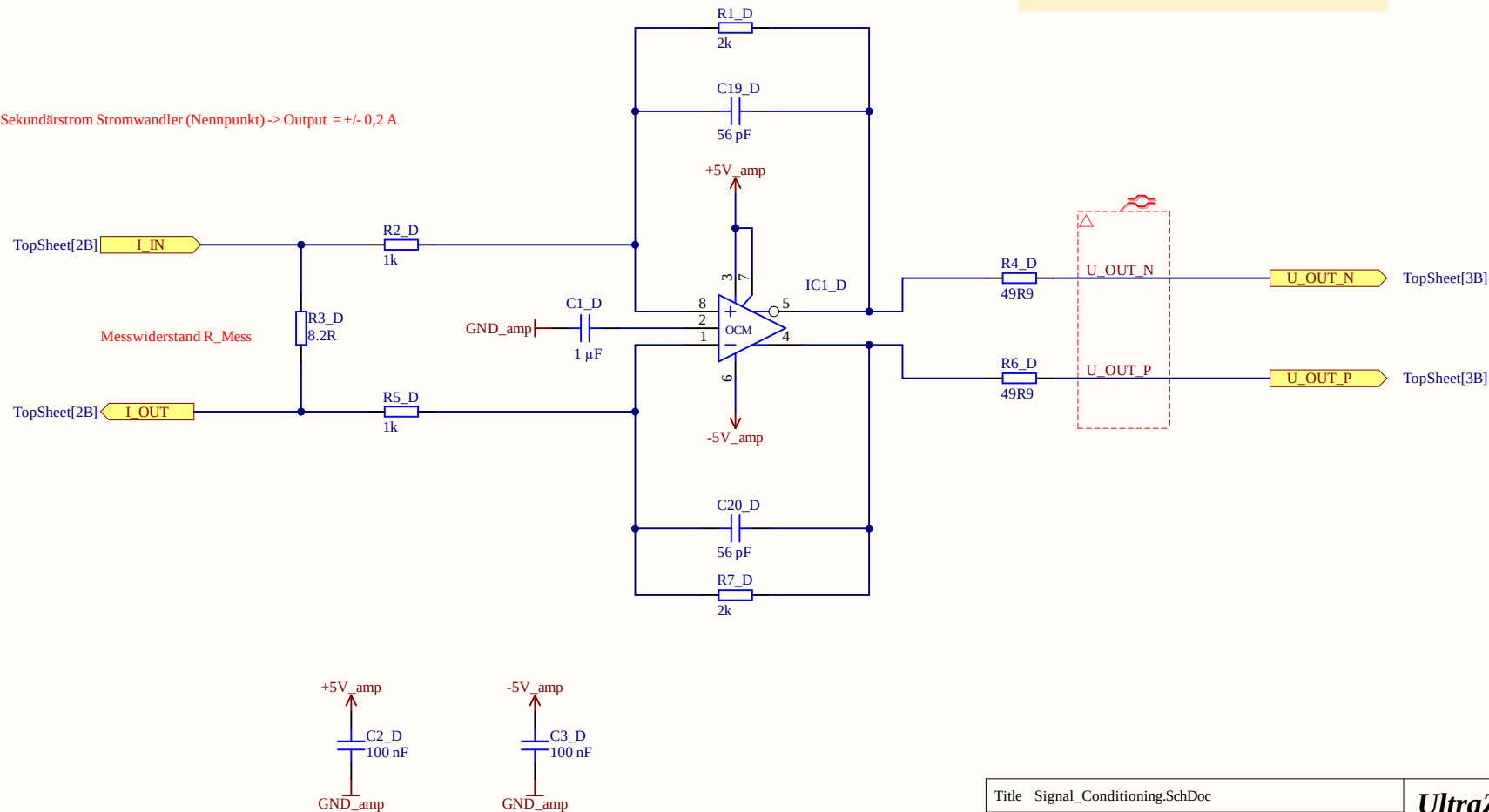
Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz: f_g

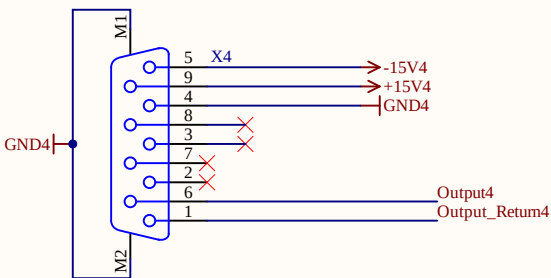
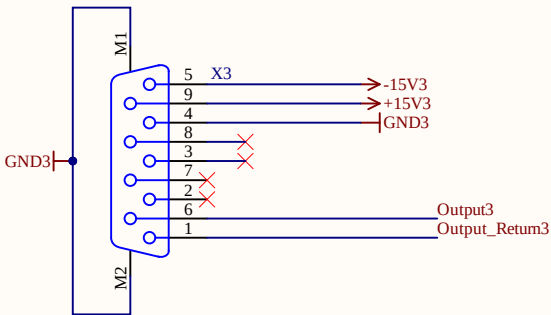
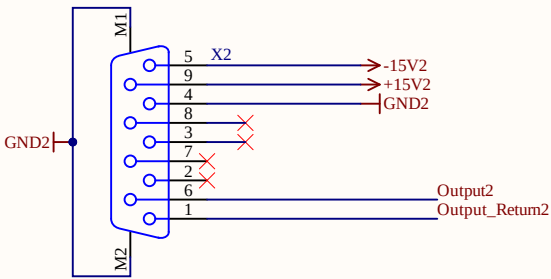
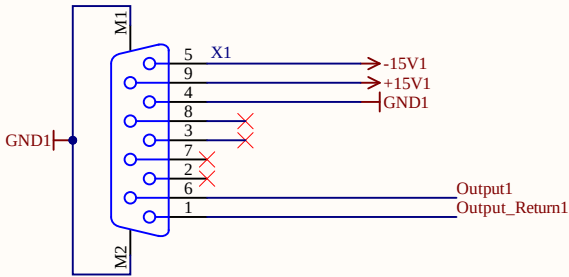
Rückkopplungskondensator: $C_f =$

$$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot C_f = \dots =$$

Sekundärstrom Stromwandler (Nennpunkt) -> Output = +/- 0,2 A



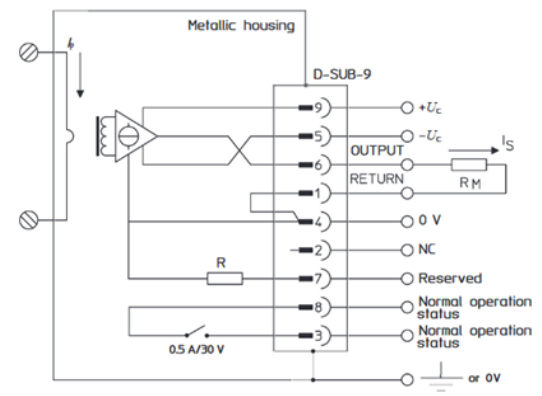
Analog Input



Connection

D-Sub connector

- 1- Output return
- 2- NC
- 3- Normal operation status
- 4- 0 V
- 5- -15 V DC
- 6- Output
- 7- Reserved
- 8- Normal operation status
- 9- +15 V DC



Output[1..4] Output[1..4] TopSheet[1B]

Output_Return[1..4] Output_Return[1..4] TopSheet[1B]

+15V[1..4] +15V[1..4] TopSheet[1B]

-15V[1..4] -15V[1..4] TopSheet[1C]

GND[1..4] GND[1..4] TopSheet[1C]

Title DSub_Analog_Input.SchDoc

Revision: 01 Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb

UltraZohm

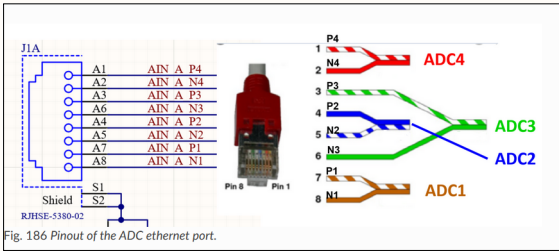
www.ultrazohm.com

Date: 27/05/2026

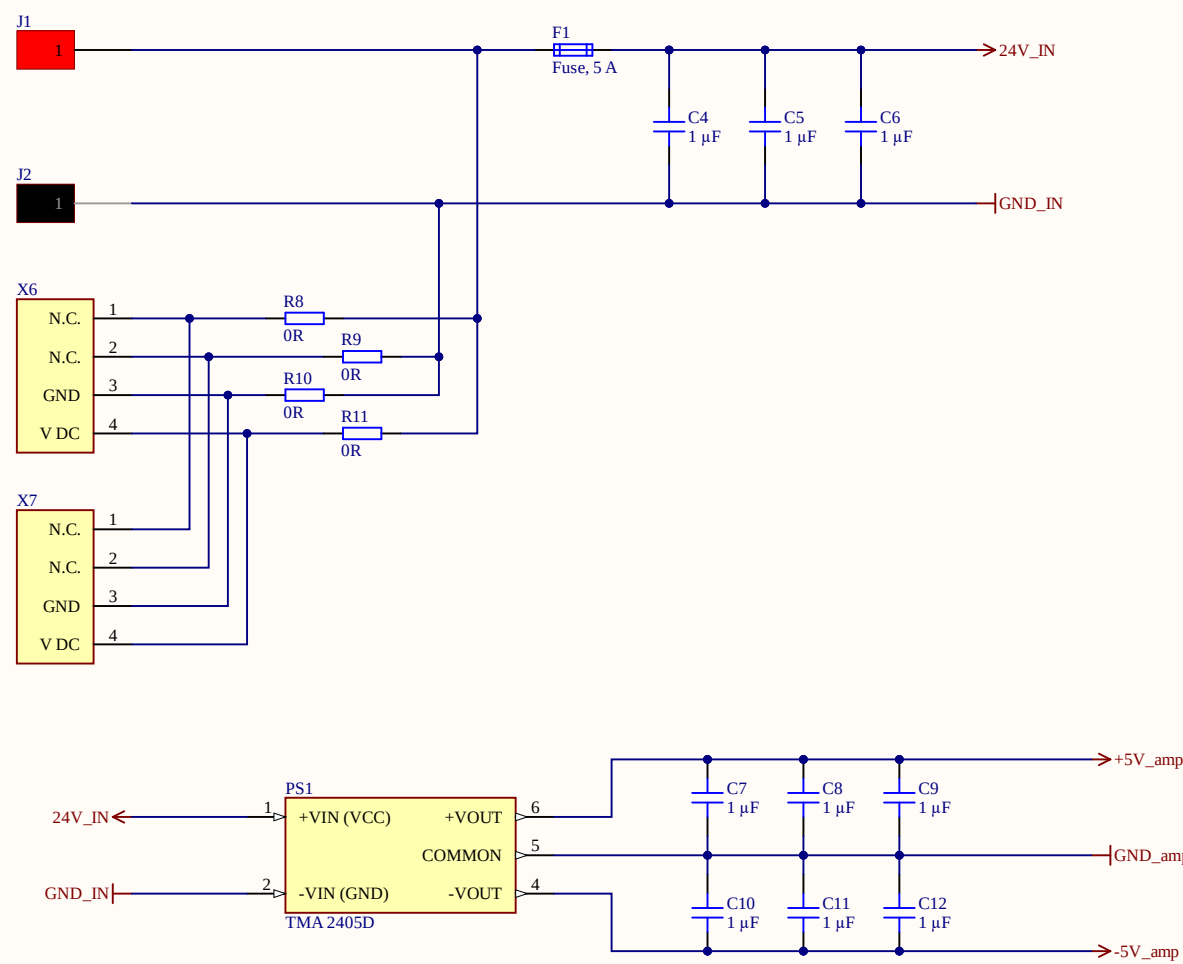
Sheet 3 of 12



Analog Output



Power Supply for Amplification Circuit



Quelle: DC/DC Wandler 24V auf +/- 5V

Verbraucher: ADA4940 fully differential Amplifier

Ruhestrom: 1,25 mA
bei 5,0 V
Power: 0,00625 mW

Output Characteristics:

U_out ~ Vs = 5 V
I_out = 46 mA peak
P_out max = 0,23 W

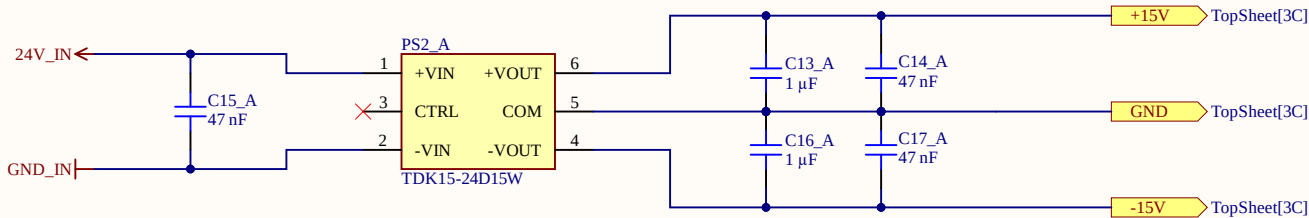
Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V
DC Nennstromaufnahmen @15V = 320 mA
I_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A
Shielded output cable & shielded plug are recommended to use

Max Power Consumption (pro Stromsensor):
P = U*I_max = 15 V * 0,420 A = 6,3 W



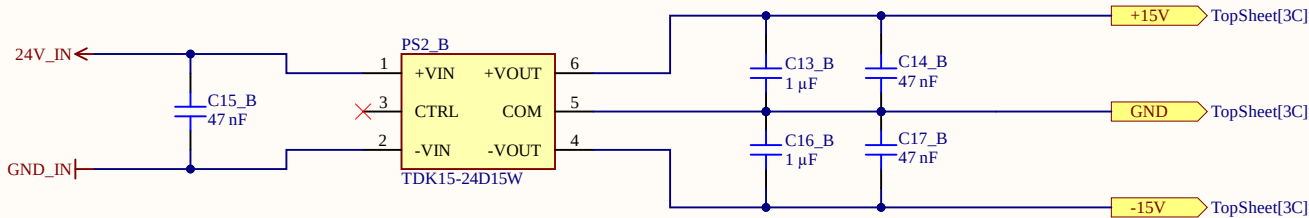
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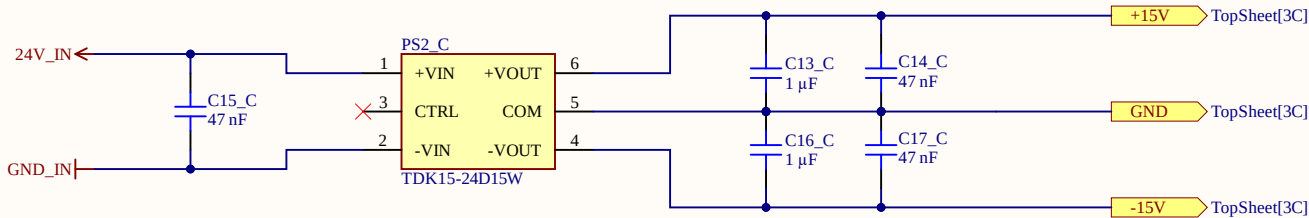
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Power Supply Currentsensors

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